Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 25)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
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6	
7	
8	
9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 State what is meant by **hybrid vigour**.

[1 mark]

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0 1 . 2 Explain how cross-breeding Zebu and Ayrshire cattle produced a breed suited to hot climates.

[2 marks]

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0 1 . 3 Give **two** potential applications of cloning to livestock farming.

[2 marks]

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2 .....

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0 2 . 1 Describe how each of the following abiotic factors can be managed to increase biodiversity in a nature reserve.

[6 marks]

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Light .....

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Dissolved oxygen .....

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Nutrients .....

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0 2 . 2 Explain how biotic factors can be managed to increase the survival of a species. Give **four** examples.

[4 marks]

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0 3 . 1 Which of the following habitats has low biodiversity but a high proportion of endemic species? Tick one box.

[1 mark]

A Tropical rainforest ☐

B Oceanic islands ☐

C Temperate broadleaf woodland ☐

D Mangroves ☐

0 3 . 2 Compare the ecological features of tropical rainforest and Antarctica that explain their very different biodiversity.

[4 marks]

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0 4 A student measured the wind speed at 10 sites inside a woodland and at 10 sites in a nearby clearing.

Table 1

Location	Mean wind speed / m s^{-1}	Standard deviation / m s^{-1}
Woodland	1.2	0.4
Clearing	3.5	1.1

0 4 . 1 Name the statistical test the student should use to find out whether the difference between the means is significant. Give **two** reasons for your choice.

[3 marks]

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0 4 . 2 Explain what the student should do if it is not possible to show that the data are normally distributed.

[1 mark]

0 4 . 3 The test gave p = 0.02. Explain what this means.

[2 marks]

0 4 . 4 Explain how the standard deviations in Table 1 support the conclusion that the means are different.

[2 marks]

0 4 . 5 Explain why a statistically significant result does not prove that trees cause the lower wind speed.

[2 marks]

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0 5 A national conservation body must decide how to divide a limited budget between species.

0 5 . 1 Describe **three** criteria the IUCN uses when deciding the Red List category of a species.

[3 marks]

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0 5 . 2 A population of lemurs has fallen from 12 000 to 2 100 mature individuals. Calculate the percentage decline.

[2 marks]

Answer ..... %

0 5 . 3 Explain **three** reasons, other than its Red List category, why one species may be given a higher conservation priority than another.

[3 marks]

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0 5 . 4 Explain the difference between CITES Appendix I and Appendix II, and explain why controlling trade can protect a species more effectively than habitat protection alone.

[3 marks]

0 5 . 5 Describe **four** ways that UK legislation and designated areas protect habitats.

[4 marks]

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0 6 . 1 Explain the difference between approaches to sustainable lifestyles based on **moral belief** and on **enlightened self-interest**.

[2 marks]

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0 6 . 2 Explain why the overall environmental impact of humans depends on two main factors.

[2 marks]

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0 6 . 3 Give **one** activity that is sustainable with a small population but unsustainable if the population is too large.

[1 mark]

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0 7 . 1 Complete Table 1 to compare natural systems with typical linear human systems.

[4 marks]

Table 1

Feature	Natural systems	Linear human systems	
Energy source			
Use of materials			
Nature of wastes			
Temperature of processes			

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0 7 . 2 Explain why natural systems can be sustained using a relatively small number of elements.

[2 marks]

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0 7 . 3 Explain why the human use of minerals makes them difficult to recover and re-use.

[2 marks]

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0 7 . 4 Explain why toxic metals in electronic products cause problems that the wastes of natural systems do not.

[2 marks]

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0 8 . 1 Explain how living organisms provide each of the following ecosystem services.

[6 marks]

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Regulation of atmospheric composition .....

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Soil maintenance .....

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Interspecies relationships .....

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0 8 . 2 Explain why Darwin's orchid of Madagascar depends on a single moth species, and why such specialisation is both an advantage and a risk.

[2 marks]

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0 8 . 3 Explain why elephants are particularly important seed dispersers.

[2 marks]

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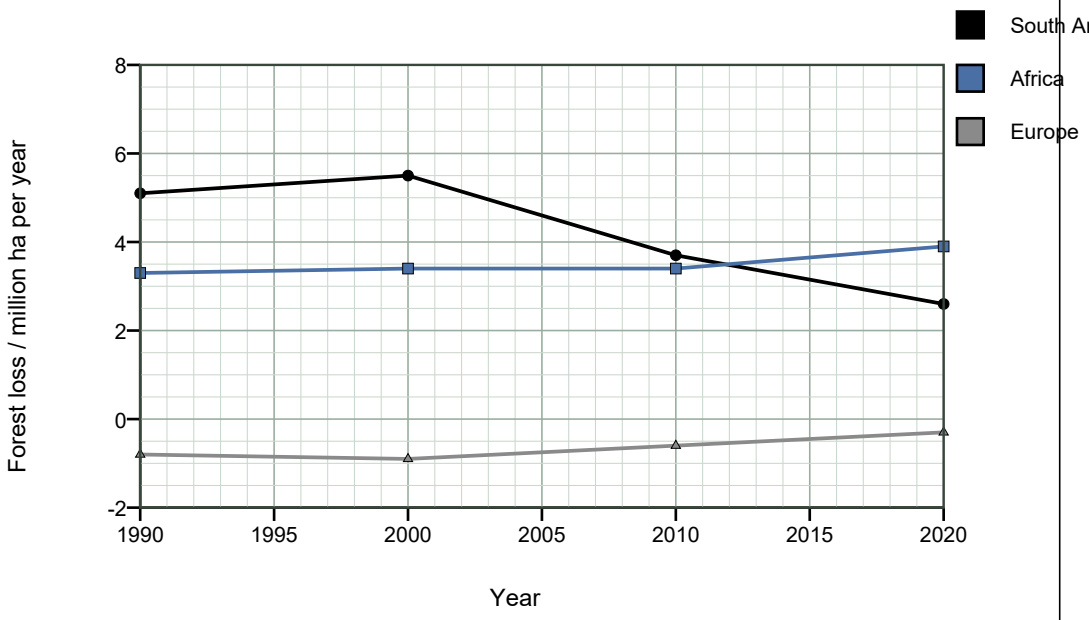
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0 9 Figure 1 shows the area of forest lost each year in three regions between 1990 and 2020.

Figure 1



0 9 . 1 Describe the trends shown in Figure 1.

[3 marks]

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0 9 . 2 Explain the main causes of forest loss in South America and Africa.

[4 marks]

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0 9 . 3 Explain why roads and railways cause more forest loss than the area they occupy.

[2 marks]

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0 9 . 4 Explain why softwood from Scandinavia and Canada is generally produced more sustainably than tropical hardwood.

[1 mark]

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1 0 Table 1 shows the number of adult great crested newts caught in bottle traps in a pond over five years, and the number of ponds in the area in which the species was detected using environmental DNA (eDNA).

Table 1

Year	Adults caught in traps	Ponds with newts detected by eDNA (out of 20)
2019	14	6
2020	22	9
2021	18	11
2022	31	13
2023	27	15

1 0 . 1 Calculate the percentage increase in the number of ponds where newts were detected between 2019 and 2023.

[2 marks]

Answer [ ] %

1 0 . 2 Explain how eDNA detects the presence of newts without catching them.

[2 marks]

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1 0 . 3 Suggest why trap catches are a less reliable measure of population change than the eDNA survey.

[2 marks]

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1 0 . 4 Evaluate the contribution of new technologies to ecological monitoring and wildlife conservation.

[9 marks]

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Write an essay on one of the following topics.

1 1 . 1 Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably.

[25 marks]

OR

1 1 . 2 Discuss how human activities influence biodiversity, using examples from a range of different contexts.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS